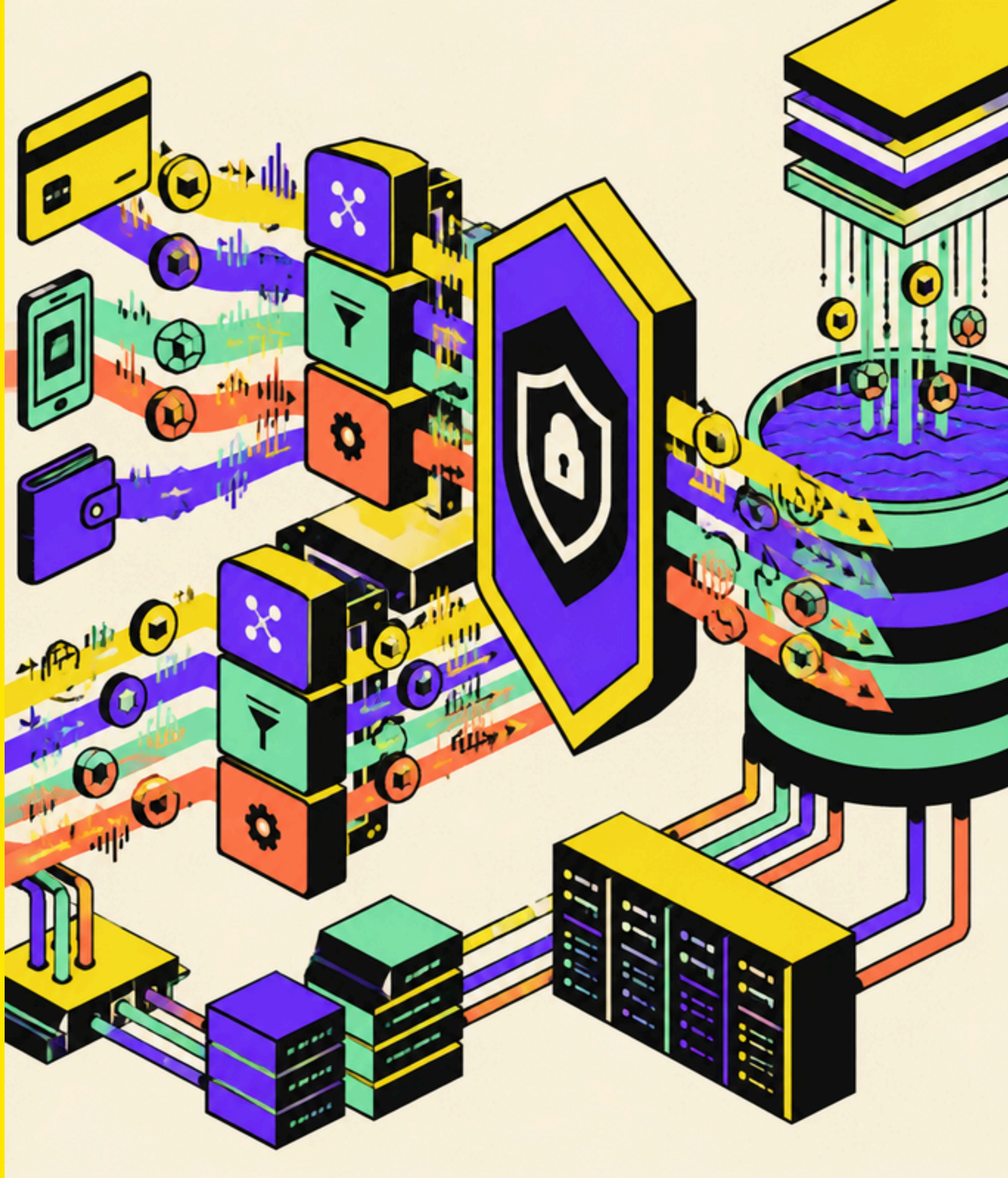
SIC8

REAL-TIME PAYMENT INGESTION

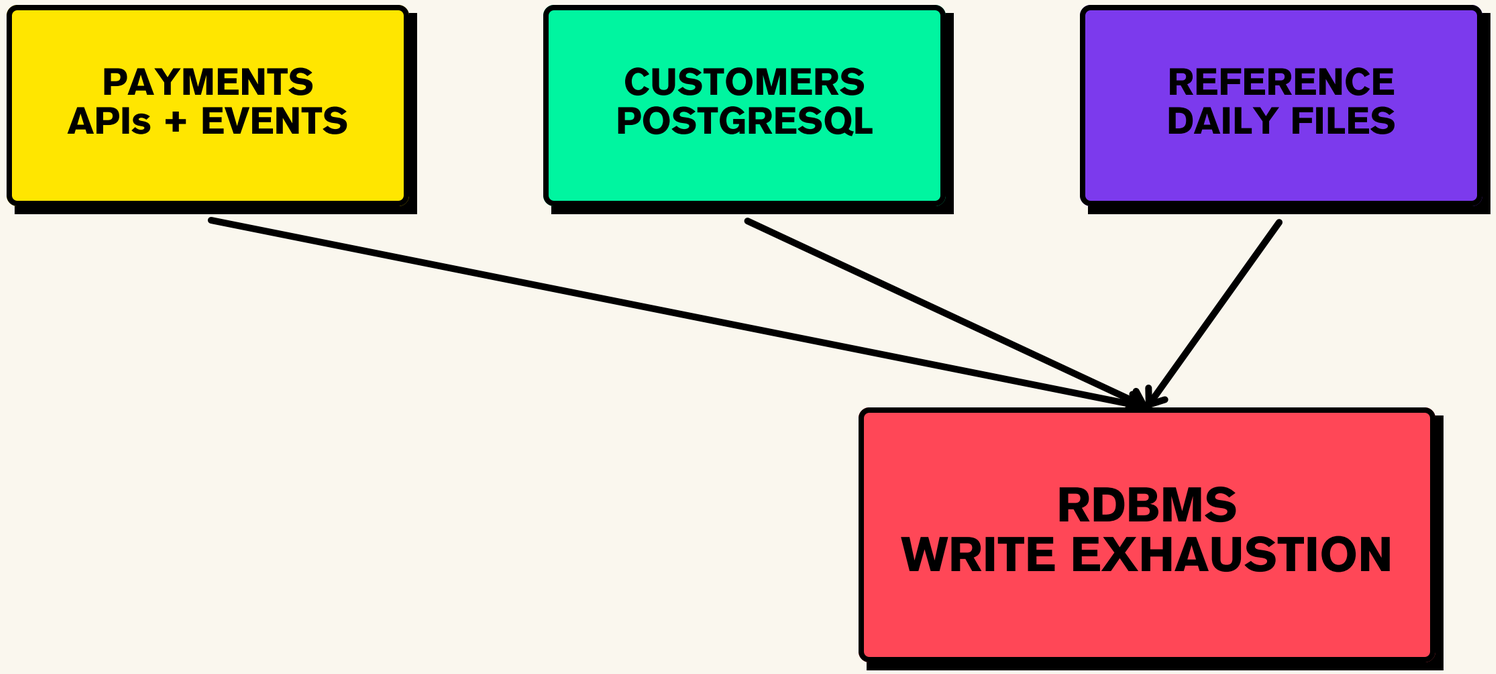
Zero DB inserts. Sub-10ms fraud scoring. One Medallion Lakehouse.

AWS S3 × DATABRICKS × MSK



THE INGESTION DILEMMA

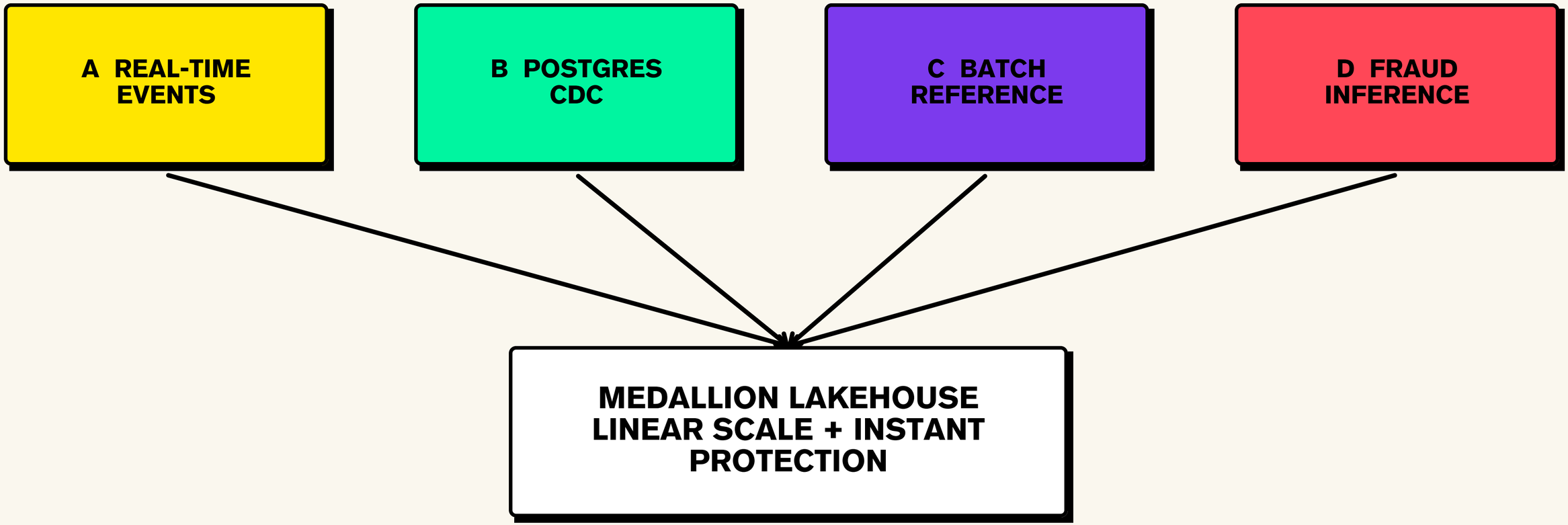
High transaction velocity turns a familiar RDBMS into a shared choke point.



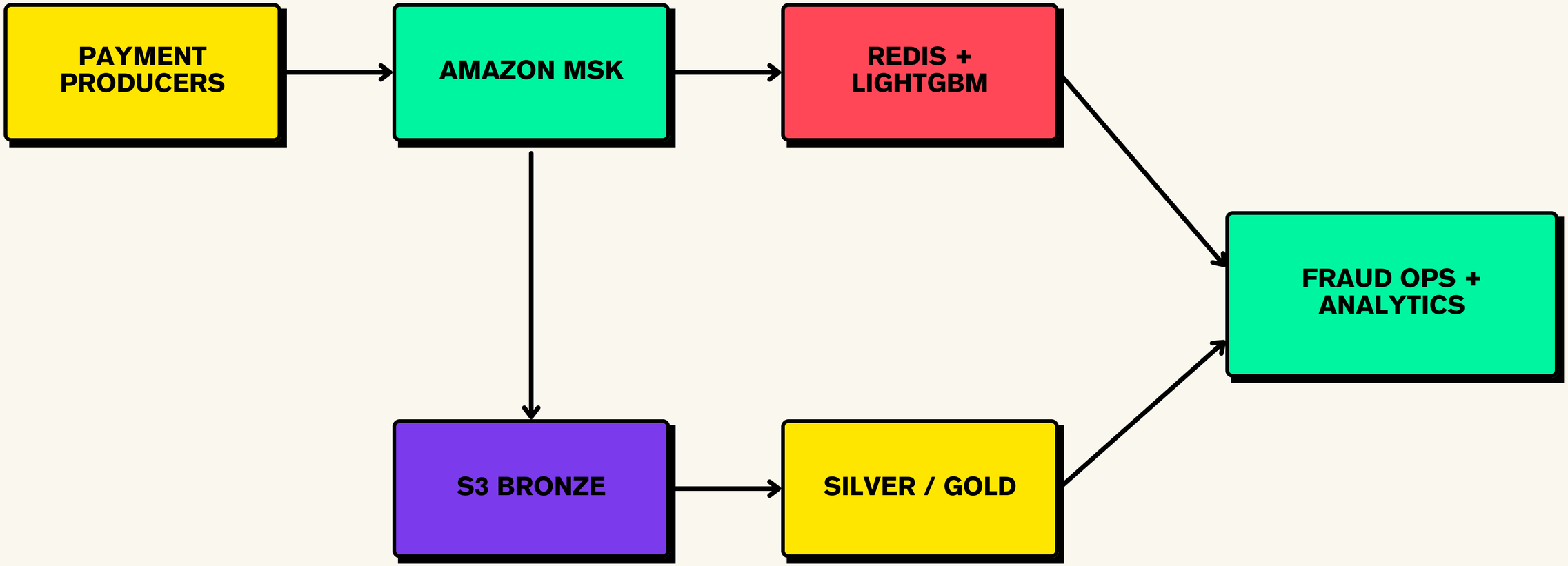
LOCKS • I/O • POLLING • DOUBLE-WRITES

DECOUPLE INGESTION FROM PERSISTENCE

Four paths, one event contract, zero relational inserts on the payment hot path.



ONE TOPOLOGY. FOUR DATA PATHS.



A — ZERO-DB STREAMING

Payment events bypass relational persistence and land directly in durable streaming infrastructure.



NO RELATIONAL WRITE ON THE HOT PATH

B — POSTGRESQL CDC

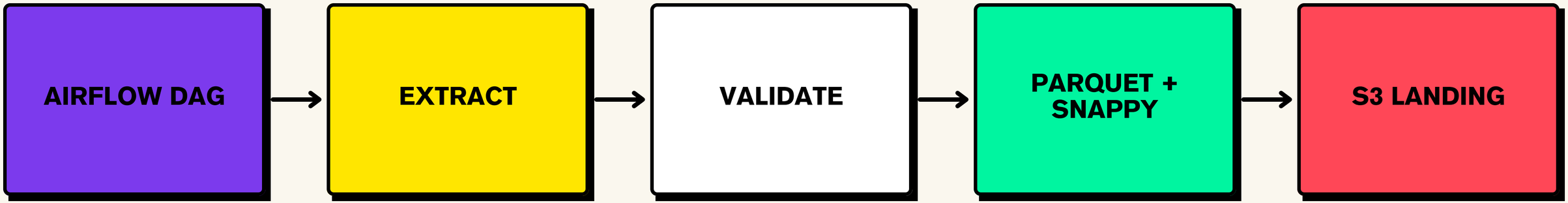
Debezium 2.5 tails WAL logs: no polling, no double-write, no application coupling.



NO RELATIONAL WRITE ON THE HOT PATH

C — SCHEDULED REFERENCE INGESTION

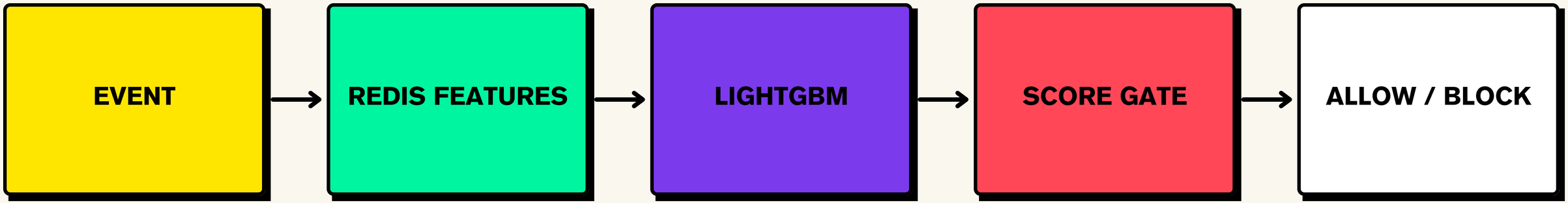
Airflow snapshots dimensions into compact, query-efficient Parquet.



NO RELATIONAL WRITE ON THE HOT PATH

D — SUB-10MS FRAUD INFERENCE

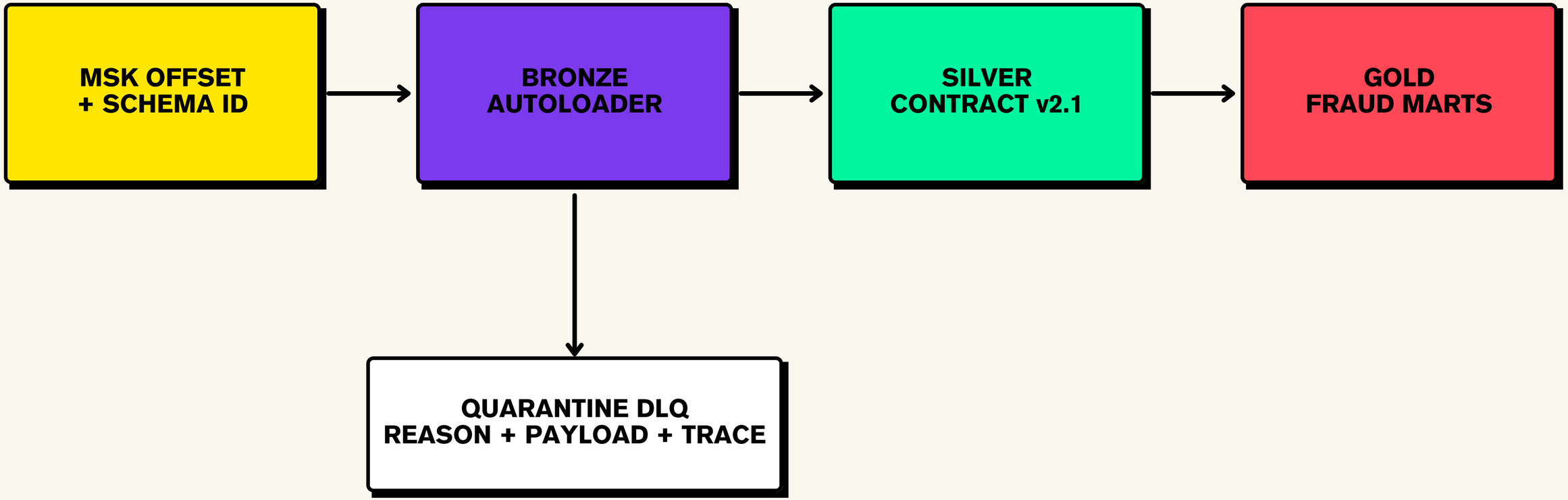
In-flight scoring hydrates behavioral state before a payment reaches authorization.



P95 TARGET < 10ms

LINEAGE IS CODE, NOT A DIAGRAM

Provenance is explicit from source offset through every Delta transformation.



MEDALLION STORAGE, BUILT FOR REPLAY

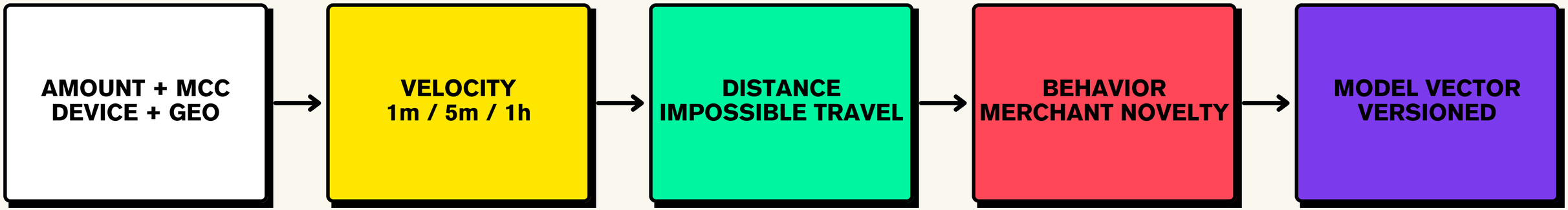
BRONZE / IMMUTABLE RAW / EVENT TIME + INGEST TIME

SILVER / CONTRACT v2.1 / DEDUPED + TYPED

GOLD / FRAUD + FINANCE MARTS

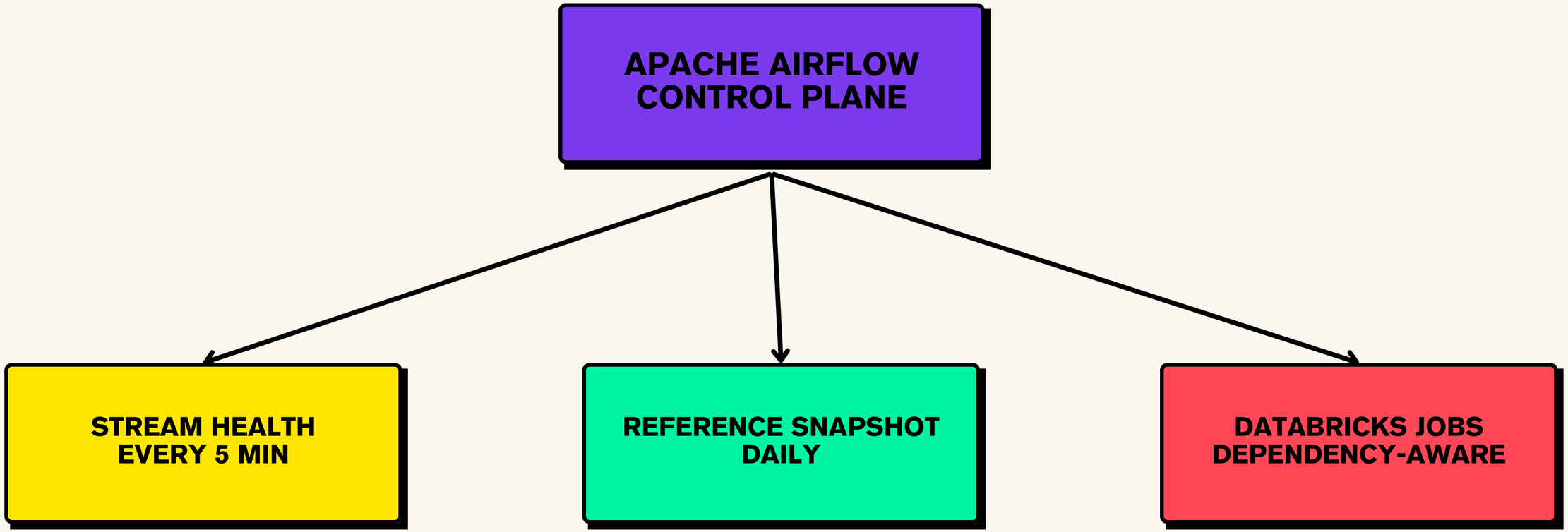
RAW SIGNAL → BEHAVIORAL INTELLIGENCE

Streaming transforms convert telemetry into high-predictive-value features in milliseconds.



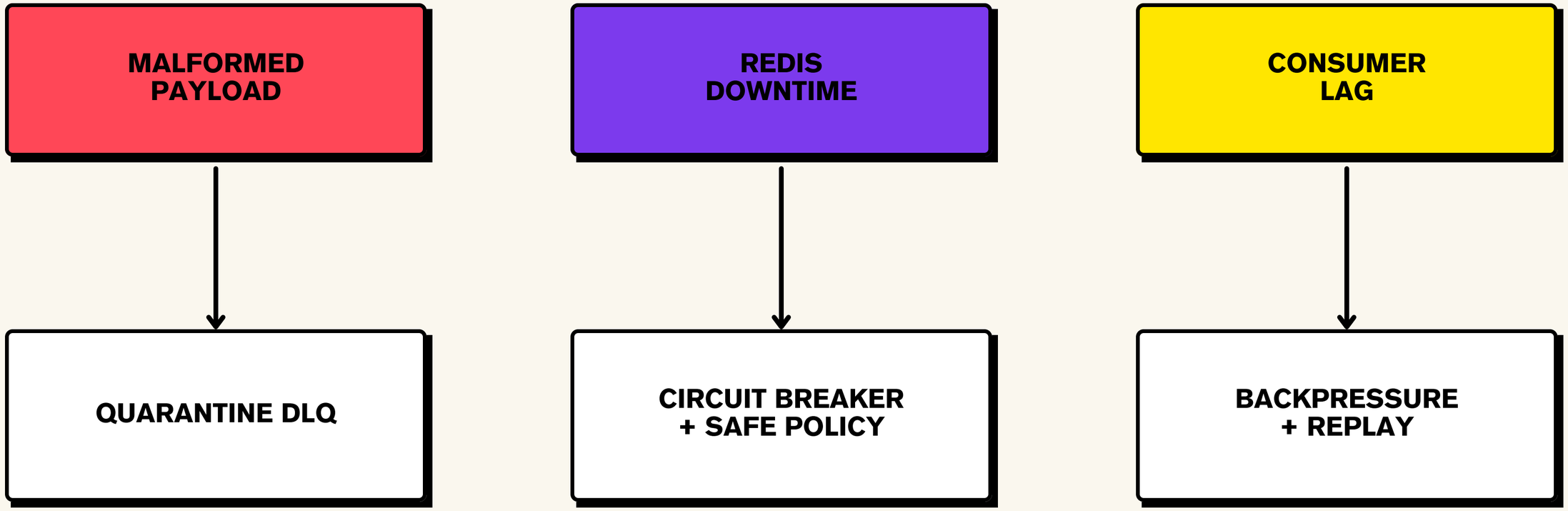
AIRFLOW CONDUCTS; STREAMING KEEPS PLAYING

Orchestration coordinates checks and jobs without becoming the payment data plane.

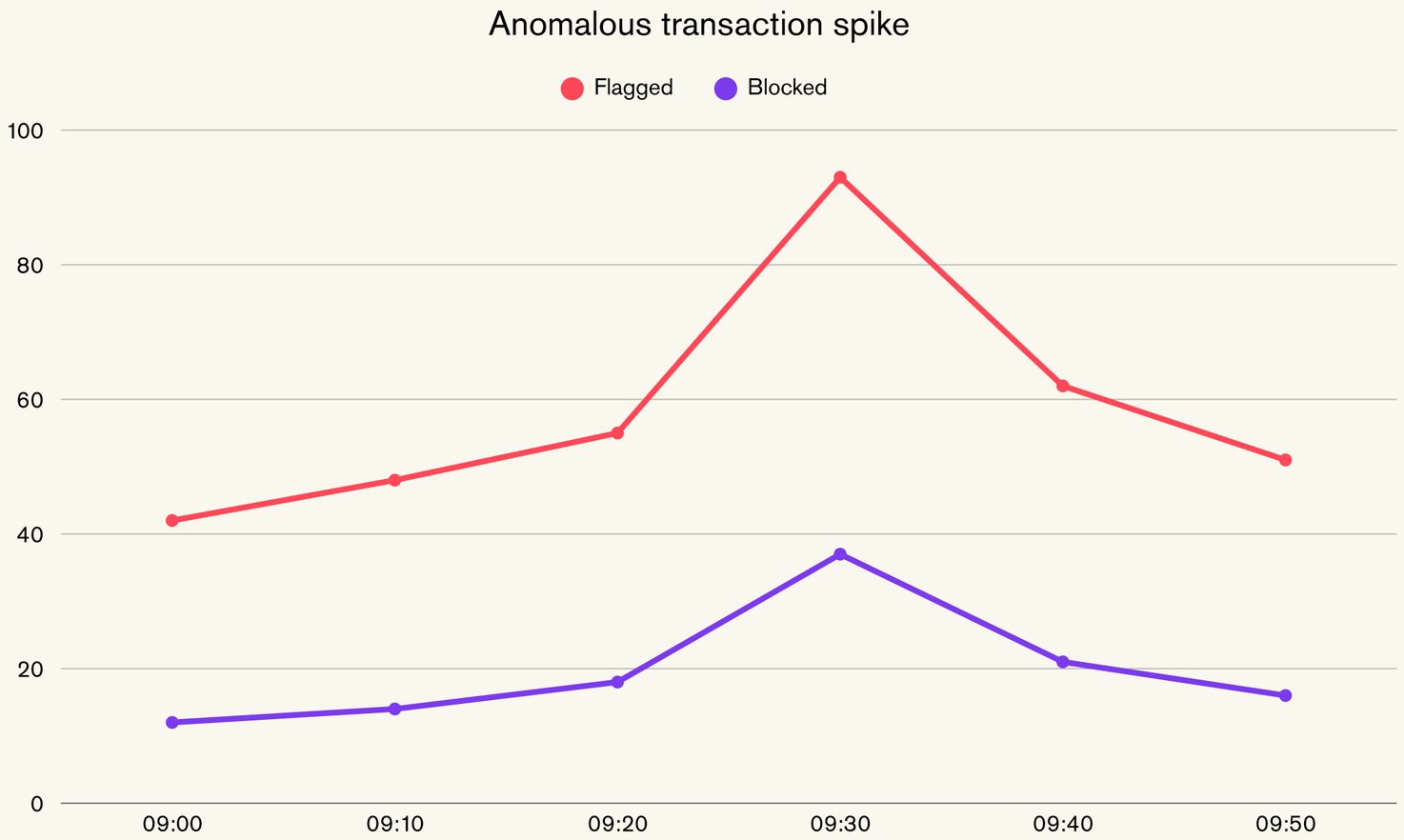


FAIL CLOSED? FAIL OPEN? FAIL TRACEABLY.

Each failure mode has an explicit containment path and recovery contract.



EXECUTIVE FRAUD OPERATIONS

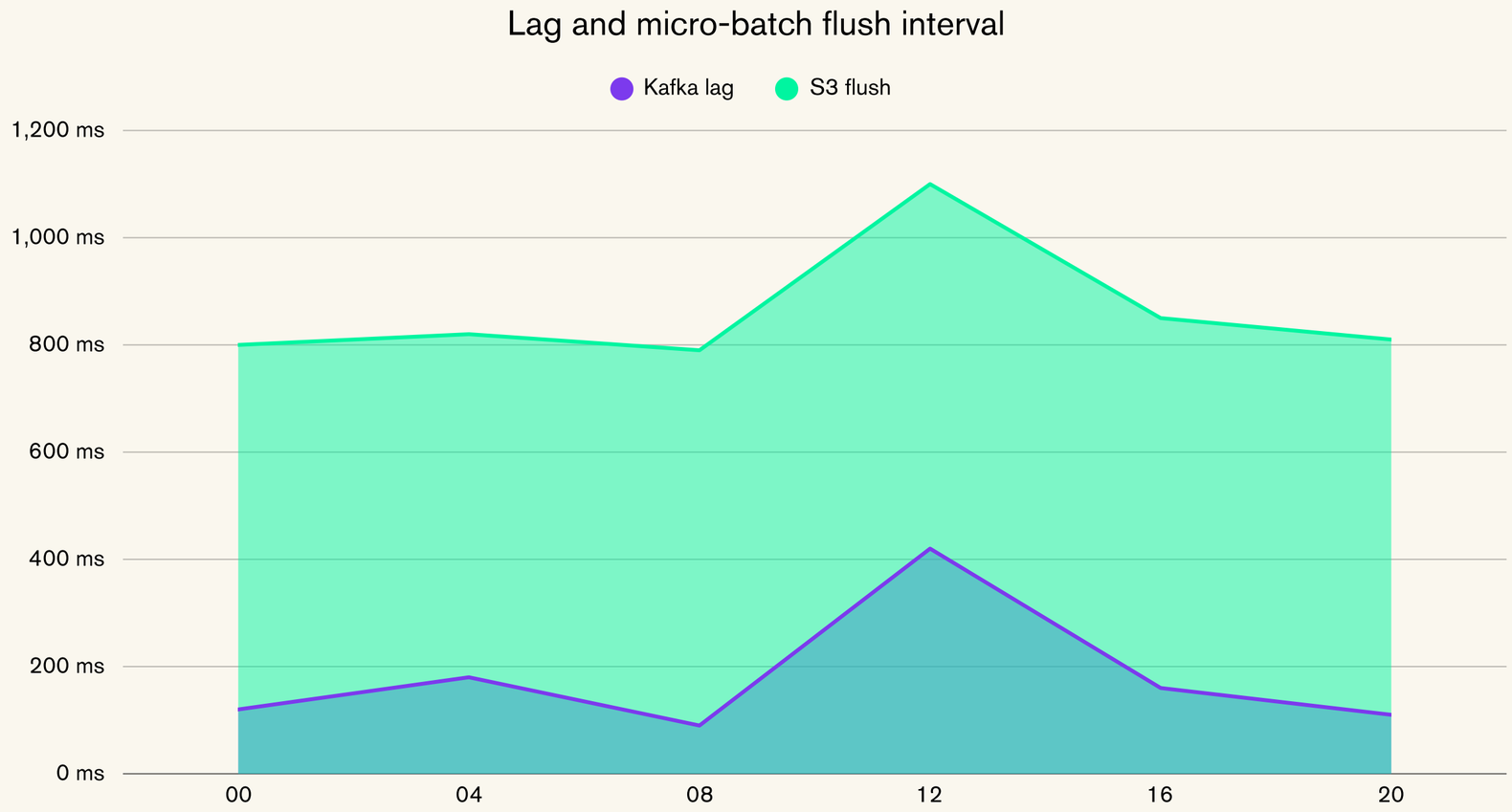


96.4%
PRECISION

$2.8M
BLOCKED

7.6ms
P95 SCORE

INGESTION & PIPELINE HEALTH

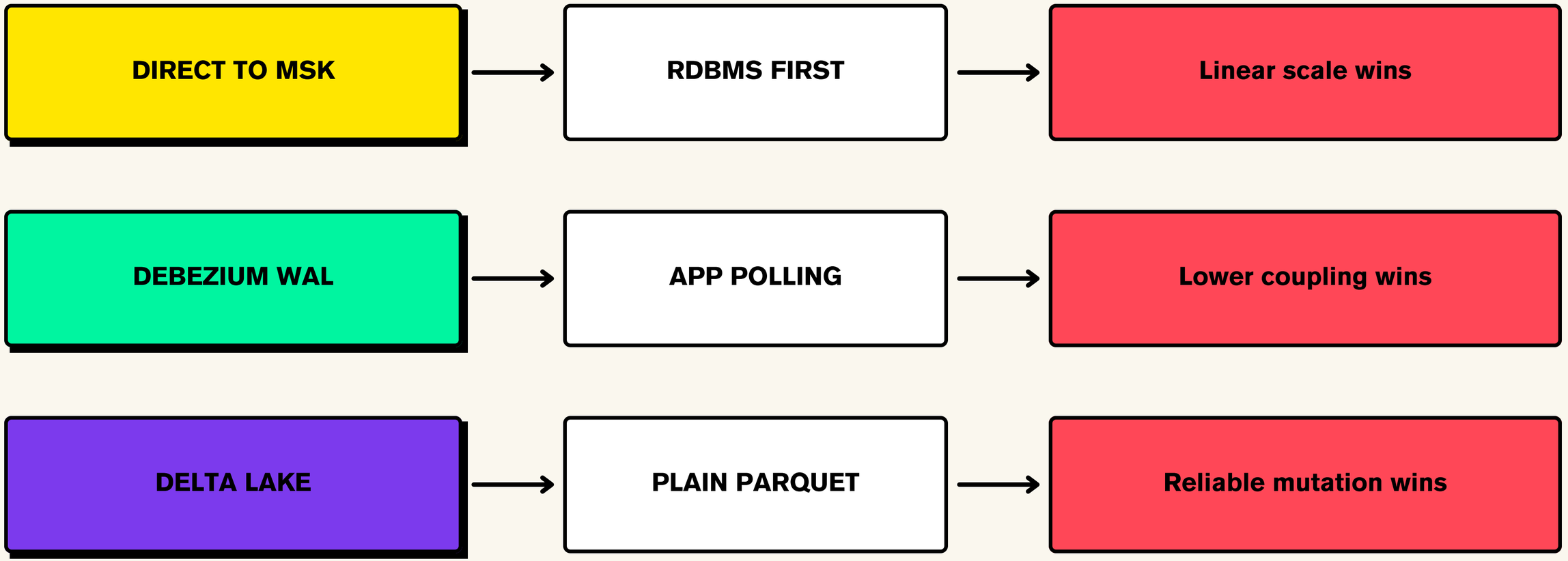


0.42s
TOPIC LAG

0.81s
S3 FLUSH

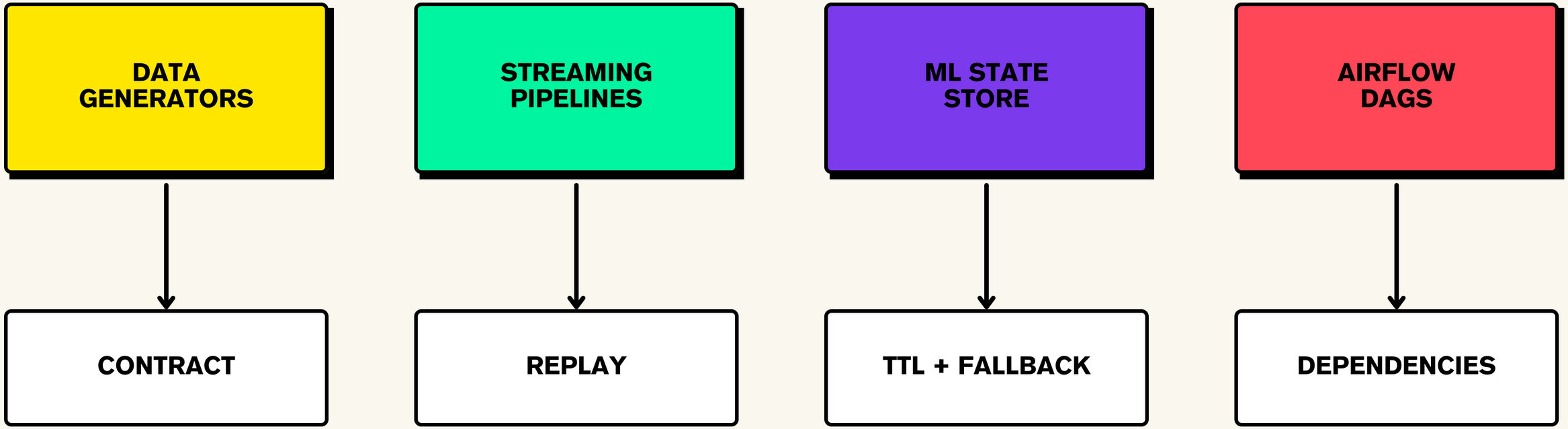
68%
CLUSTER

DELIBERATE TRADE-OFFS



TEST THE SYSTEM, NOT JUST THE FUNCTIONS

Parity across generators, transport, state, transformations and orchestration.



UNIT → INTEGRATION → CHAOS → RECOVERY

SHIPPED FOR SCALE. READY FOR HARDENING.

RESULTS

Zero DB inserts
Sub-10ms scoring
Replayable lineage

LIMITS

Single-region baseline
Feature freshness risk
Manual model promotion

NEXT

Multi-region MSK
Online feature store
Automated model gates

SIC8 turns payment velocity into a competitive advantage.